

Unstable Bluff Detection System

Dominic Ricci

Ryder Rochoa

Dylan Hancock

Alex Hardy

Project Description

Task:

The Unstable Bluff Detection System is a system to detect changes in the bluff holding up Del Mar, California's train tracks. If the bluff appears to be shifting or falling apart, the system will send an alarm to the proper authorities. The system serves as a warning system for potentially hazardous conditions, which could endanger passengers and cause damage to the railway.

Logistics:

The system works via 6 fixed cameras placed along the bluff. The cameras are given power and a Wi-Fi connection to a central server. The server contains imaging software and a database to store images up to a month old. As well as other images, once per day, up to a year old. According to the project description, the imaging software and database are externally handled.

Design:

For our part of the ecosystem, we designed two systems: The camera system and an interface system for the server. The camera system is present on each of the 6 cameras. It determines how the camera operates. It also sends information to the interface system. The interface system communicates with the already-handled imaging software and database.

Camera System

Externally, a time interval is set. Then, every time interval, the camera will take a photo. The camera will also collect a timestamp from the internet and its geo-locational coordinates from GPS. Then, it will send the image over the network to our server-side interface. As per the project description, the timestamp and locational data are stored in the image.

Server-Side Interface System

On the server side, we have three components: CameraListener, CameraConfig, and AlarmSystem. CameraListener waits for the camera to send image data. From there, the listener sends the data to the image software. The image software will then store the image in its database and determine whether it should send information about the image to the AlarmSystem. The AlarmSystem simply broadcasts information about the image to the authorities.

The CameraConfig determines the camera's resolution, time interval, and whether it takes pictures in color or greyscale. When a change is manually made to the config, it sends the updated information over the network to the camera. The camera then updates its local config based on this information.